Meet the 38: Interfaceness

The Active, Constitutive Boundary as an Ontological Necessity

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Abstract

This paper introduces Property 38 — Interfaceness — within the framework of Superreality and the Property Method. Building on the axioms ChOR $\to \infty$, KSS $\to \infty$, and PPU $\to \infty$, we argue that boundaries between ontological regimes (worlds) cannot be reduced either to objects of one world or to mere relations, but must be recognized as active, constitutive zones of translation and transformation. Interfaceness is defined as the capacity of a boundary to function as a domain where patterns are transcoded across different ontological media. We distinguish two inseparable aspects: the inter-aspect (connection between distinct worlds) and the aut-aspect (the boundary as both inner alterity and external point of contact). The paper differentiates Property 38 from the Γ-operator (Property 34), clarifies its relation to the world of boundaries (W_4), and establishes that interfaceness is not binary but exhibits graded intensity measurable through Salience (P37), Bindability (P34), and Propertyness (P36). Examples from physics (event horizons, quantum vacuum), biology (cell membranes), mathematics (zero), and consciousness studies demonstrate the universality of the principle across both inter-world and intra-world transitions. Property 38 addresses a systematic blind spot in describing the process of transition itself, completing the system of 37 properties without ontological circularity or duplication.

Keywords: interfaceness, ontology, boundaries, Superreality, Property Method, translation, transformation, event horizon, consciousness, world of boundaries, Γ-operator, connectivity, phase transition

1 Foundation

The necessity for a new Property (within the framework of Superreality and the Property Method) emerges from the combined application of the axioms **ChOR** $\to \infty$, **KSS** $\to \infty$,

and **PPU** $\to \infty$. In analyzing phenomena where the boundary between worlds (between W_1 and W_2, W_1 and W_3, W_2 and W_3) cannot be reduced either to an object of one world or to a mere relation, the three axioms collectively necessitate the recognition of the *active, transformative character of the boundary.* Every boundary is ontologically rooted in the possibility of actualization within the world of boundaries (W_4); however, Property 38 itself designates not W_4 itself, but the capacity of a *concrete* boundary to serve as a zone/domain of translation. This requirement *is* Property 38. It is not derived from anything more fundamental but is discovered as an invariant without which the description of such phenomena remains incomplete or contradictory.

2 Definition: Interfaceness as the Dialectic of the "Between," the "Self," and the "Outer"

Interfaceness is the capacity of a boundary between ontological regimes (worlds) to function not as a passive divider, but as an *active, constitutive zone/domain of translation, transformation, and mutual determination.* The term derives from the Latin *inter* ("between"), yet its conceptual scope encompasses three inseparable aspects.

Inter-Aspect — the connection between distinct ontological regimes. The interface links W_1 and W_2, W_1 and W_3, W_2 and W_3, enabling the translation of patterns from one authentic medium into another. Without this aspect, the interface could not fulfill its transitional function.

Aut-Aspect — an intentionally polysemous term that fuses two meanings. From the Greek *autos* ("self") — the interface's capacity to be the *inner otherness of a world*, its constitutive boundary, that which a world carries within itself as alterity (e.g., the event horizon as the intrinsic boundary of a black hole). From the English *outer* — the interface's capacity to be a *point of external connectivity*, a site of encounter with another world (e.g., the event horizon as the zone of interaction with infalling matter). Within the interface, these two meanings converge: the boundary is simultaneously the innermost (constitutive for the world) and the outermost (the point of contact with the other). Without *autos*, the interface would be a mere passage between ready-made worlds; without *outer*, it would be a windowless monad.

In the limiting case where the interface operates within a single ontological regime (e.g., the water–ice phase transition), both aspects merge, giving rise to the **auto-interface** — a boundary between states of one and the same medium, which simultaneously distinguishes and connects those states. Such an auto-interface does not generate a new world (unlike W_4), yet it demonstrates the universality of the principle of interfaceness.

Distinguishing the Γ-Operator from Property 38. The Γ-operator (Property 34) describes a *singular act of transition* from potentiality (P25) to onticity (P33). Interfaceness,

by contrast, designates the *capacity of a boundary to be the site* where such acts are possible repeatedly, with varying intensity and across different worlds. Γ is an event; P38 is a property of the boundary that enables the event to occur. Without P38, the Γ-operator would be an act in a void, devoid of a locus for translation. Introducing P38 does not duplicate Γ but complements it, establishing the ontological status of the boundary as the condition of possibility for Γ-acts.

Interfaceness is thus reducible neither to a pure "between," nor to a pure "interiority," nor to a pure "exteriority." It is a mode of being of the boundary that simultaneously distinguishes and connects, that is external to the internal and internal to the external. The interface possesses no ontological substrate independent of the worlds it joins; its being is exhausted in the act of transition, and it is precisely in this sense that it belongs to the world of boundaries, W_4. It is polyvalent (capable of linking multiple worlds simultaneously) and exists only insofar as it sustains the paradoxical tension (PPU $\to \infty$) between the sides it connects.

3 The Measure of Interfaceness: Graded Intensity

Under the axiom **KSS** $\to \infty$ (absolute connectivity), any boundary is ontologically already an interface, for it is a relation. The question is not the *presence* of the property, but the *intensity of its manifestation.* Interfaceness is not binary; it is always present but can vary from the arbitrarily small (an almost indiscernible, "dormant" interface) to the maximally intense (full actualization). A "passive boundary" is not the absence of an interface, but an interface whose intensity tends toward zero, falling below the threshold of detectability or phenomenological salience. This understanding eliminates the category mistake of conflating quality and quantity: the property itself (being an interface) is graded, not binary, which corresponds to the continuous character of connectivity in Superreality.

The intensity of an interface's operation is scale-relative and assessed through other properties:

- **Salience (P37)** — the measure of transformative potential, the threshold of transition actualization.
- **Bindability (P34)** — the number and stability of connections the interface can sustain simultaneously.
- **Propertyness (P36)** — the complexity of the pattern translated through the interface.

Examples of low intensity (the interface exists but operates weakly):

- A weak quantum interaction between two particles that rarely exchange virtual photons.

- Weak gravitational coupling between distant galaxies.
- The boundary between oil and water at rest: surface tension exists, yet material exchange is minimal.
- A weak emotional bond between casual acquaintances — an interface of consciousnesses exists, but the translation of states is nearly zero.

Examples of high intensity (tending toward the limit):

- The event horizon of a supermassive black hole in an active accretion phase: colossal transformation of mass-energy into radiation.
- The moment of creative insight: an instantaneous reconfiguration of the cognitive system, generating a new, complex pattern.
- The quantum leap of an electron between orbitals in an atom — a discrete, indivisible act of transition whose intensity is determined by the energy difference and represents the maximum possible for that system.

4 Relation to the World of Boundaries (W_4) and the Axiom KSS $\to \infty$

To avoid an ontological circle, it is necessary to distinguish:

- **W_4 (the world of boundaries)** — an autonomous ontological regime whose ontological fabric consists of acts of transition, interpretations, Γ-events. It is the world in which transformations between other worlds are possible. W_4 is not composed of interfaces but is the ontological medium that makes interfaces possible.
- **Property 38** — the capacity of a concrete boundary to be a zone/domain of translation. It neither replaces W_4 nor constitutes a law of W_4; it is a local manifestation of the fact that a boundary actualizes processes characteristic of W_4.

The axiom **KSS** $\to \infty$ asserts the possibility of connection between any entities whatsoever. Yet this possibility does not entail the automatic actualization of a qualitative transition between worlds. Non-locality (P11) describes correlations independent of spatial distance, but it does not describe the transformation of a pattern in the passage from W_1 to W_2. Interfaceness designates precisely the *active, constitutive character of the boundary*, where what occurs is not mere correlation but transcoding. KSS $\to \infty$ provides potential connectivity; P38 is the mechanism of *selective* actualization of this connectivity at a concrete boundary. Without P38, connectivity would remain purely potential. Thus, P38 does not contradict KSS $\to \infty$ but is its operational complement.

5 Additional Examples

Black Hole. The event horizon belongs neither to the material world (W_1) nor to the singularity. It is a zone where matter is transcoded into radiation, and back-reaction (Hawking radiation) transforms the boundary itself. Here the inter-aspect (the connection of W_1 with the singularity) and the aut-aspect (the horizon as the black hole's intrinsic boundary and simultaneously the external interface for infalling matter) are inseparable. This is an example of an inter-world interface of high intensity.

Consciousness. Conscious experience links neural dynamics (W_1), semantic structures (W_2), and phenomenal qualities (W_3). Consciousness is neither an object of W_1 nor a relation between W_1 and W_2 — it is an interface through which these three spheres are simultaneously actualized at the locus of the "I". The inter-aspect manifests as the linkage of three worlds; the aut-aspect manifests as the inner otherness of each of them with respect to consciousness (a neuron is not an experience, yet experience is impossible without neurons). This, too, is an inter-world interface.

Cell Membrane. The membrane separates the internal milieu from the external, but not passively. It mediates selective transport and sustains a gradient. Without this tension, the membrane ceases to be a membrane. The interface here is constitutive for the object itself: the aut-aspect (the membrane as the cell's intrinsic boundary) and the inter-aspect (the connection between inside and outside) are identical. This is an example of an auto-interface within W_1 (the physical world), where the interface operates between states of one and the same ontological medium.

Mathematical Zero. Zero is simultaneously a number (the neutral element) and the absence of number (the prohibition of division), the boundary between positive and negative values. It is not an object alongside other numbers — it is an interface that organizes the number line as a structure of distinctions. The aut-aspect (zero as the intrinsic boundary of the number line) is here primary with respect to the inter-aspect (the connection of the positive and negative branches), yet the two do not exist without each other. This is an example of an auto-interface within W_2 (the semantic world). Such an intra-world interface does not generate a new world (unlike W_4) but is a local configuration within a single ChOR. It does not negate the specificity of inter-world boundaries but demonstrates the universality of the principle of interfaceness.

Vacuum. The vacuum in quantum field theory is not emptiness. It is a field in which virtual particles are continuously created and annihilated. The vacuum is an interface between potentiality and actuality, between "nothing" and "something." Its aut-aspect (the vacuum as the intrinsic boundary of the field itself) and its inter-aspect (the connection between the virtual and the actual) converge in a single process: the creation of a particle constitutes the actualization of the interface. This, too, is an auto-interface within W_1, but at the quantum level.

6 Status

Property 38 is not a meta-property, nor does it annul other properties. It designates what had remained unaddressed: the *process of transition itself*, which can be described neither as an object nor as a relation between objects. Interfaceness is introduced to remedy a systematic blind spot that arises when one attempts to describe a boundary solely in terms of what it joins. Interfaceness does not introduce a new entity but designates a *mode of operation of the boundary* that had previously remained unmanifest. Distinguishing the inter- and aut-aspects (the latter as the unity of *autos* and *outer*) permits the property to be applied both to inter-world and to intra-world transitions, without reducing interfaceness either to pure connectivity or to pure boundedness.

All the distinctions introduced — between the Γ-operator and P38, between W_4 and interfaceness as a local property, between inter-world and intra-world interfaceness, as well as the graded nature of intensity — are not concessions to criticism but necessary refinements that follow from the combined application of the axioms **ChOR** $\to \infty$, **KSS** $\to \infty$, and **PPU** $\to \infty$. Property 38 organically completes the system of 37 properties, introducing neither duplication nor ontological circularity.

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This study was conducted using the conceptual apparatus of the Ontology Lab (MPO System), where the properties of Superreality and axioms (ChOR $\to \infty$, KSS $\to \infty$, PPU $\to \infty$) served as implicit coordinates for navigating the topic. In the final text, we consciously avoid this terminology so that the reader may focus on the heart of the matter. For those interested, a comprehensive description of the method is available in public sources.